

Dual slope ADC

Dual Slope converter:

Adv: It requires small amount of circuit components. We can use many levels for quantization.

Disadvantage: Relatively slow.

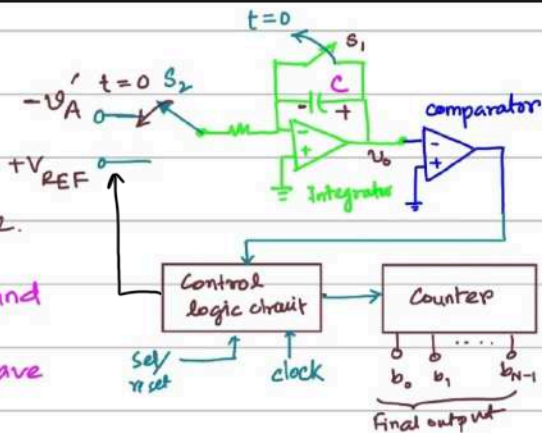
Circuit:

V_A' = Sampled and held signal

V_{REF} = preassigned value.

☐ ($t < 0$) S_1 is closed and S_2 is open.

Capacitor doesn't have any charge.



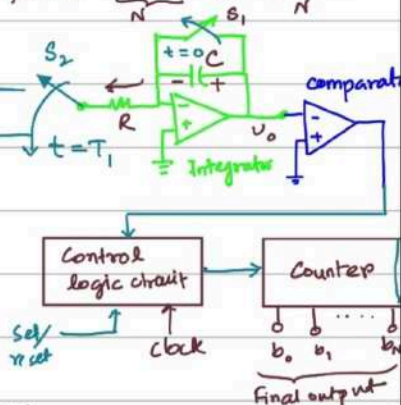
☐ $t = 0$, S_1 is open. S_2 is now connected to V_A' .

Capacitor is charging in a way that V_0 increases. Counter starts counting from $000...0$ to $111...1$.

☐ $t \geq T_1$. Counter finishes counting from 0 to $2^N - 1$.

at $t = T_1$, S_2 is now connected to V_{REF} .

From then capacitor discharges and V_0 starts to decline.



☐ $t = T_2$ (not fixed time)

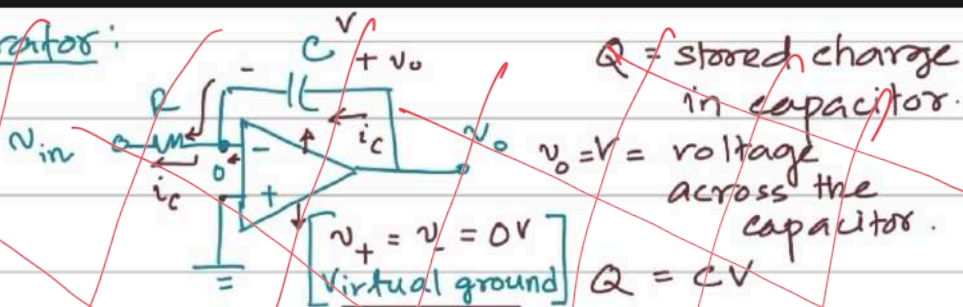
V_0 output level decreases below 0V.

Now comparator output changes from low to high.

At that moment control circuit stops the counter.

Then the output of the counter is the digital output we want.

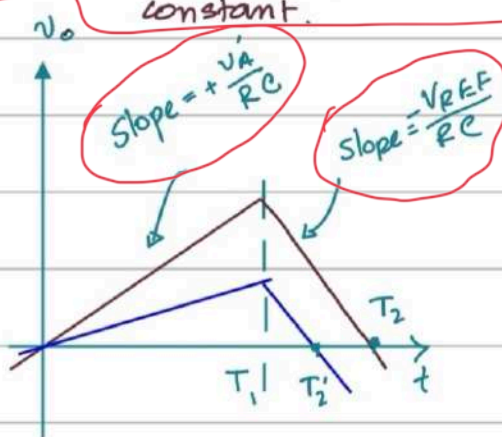
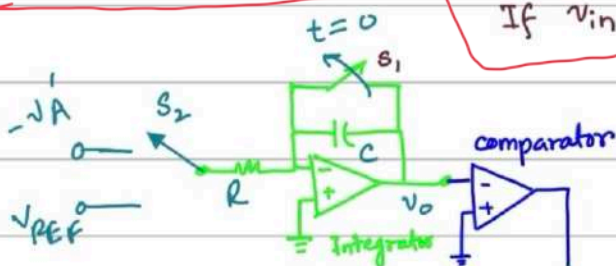
Integrator:



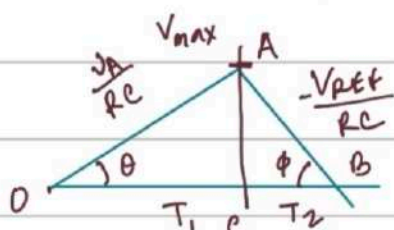
$\therefore \frac{dQ}{dt} = \text{current through the capacitor}$

$$= C \cdot \frac{dV_o}{dt} = i_c = \frac{0 - v_{in}}{R} \Rightarrow \boxed{\frac{dV_o}{dt} = -\frac{v_{in}}{RC}}$$

If v_{in} is constant then $\frac{dV_o}{dt}$ is constant.



Say final output is binary representation of the number n .



$$\tan \theta = \frac{V_A}{RC} = \frac{AC}{OC} = \frac{V_{max}}{T_1}$$

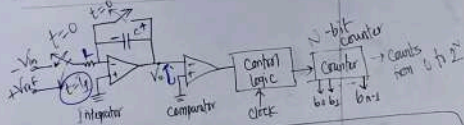
$$\tan \phi = \frac{V_{REF}}{RC} = \frac{AC}{BC} = \frac{V_{max}}{T_2}$$

$$\frac{V_{max}/T_1}{V_{max}/T_2} = \frac{V_A}{V_{REF}} = \frac{T_2}{T_1} = \frac{nT}{2^N T}$$

$$\boxed{n = V_A \frac{2^N}{V_{REF}}} \checkmark \leftarrow$$

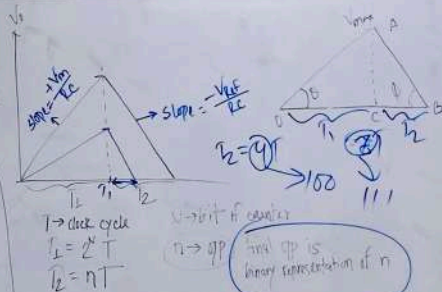
$$\boxed{V_A = \frac{n}{2^N} V_{REF}} \checkmark$$

Dual Slope ADC



$V_{in} \rightarrow$ input sampled signal
 $V_{ref} \rightarrow$ pre-assigned value

i/p voltage range = $(V_{ref} - 0)$
 number of steps = 2^N
 resolution/step size = $\frac{V_{ref} - 0}{2^N}$
 total reading time = $T_1 + T_2$



$$\tan \theta = \frac{V_{in}}{RC} = \frac{AC}{BC} = \frac{V_{max}}{T_1} \quad (I)$$

$$\tan \theta = \frac{V_{ref}}{RC} = \frac{AC}{BC} = \frac{V_{max}}{T_2} \quad (II)$$

$$(I) \div (II)$$

$$\frac{V_{in} T_1}{V_{ref} T_2} = \frac{V_{in}}{V_{ref}} = \frac{T_2}{T_1} = \frac{n T}{2^N T}$$

$$i/p: \boxed{V_{in} = \frac{n}{2^N} V_{ref}}$$

$$o/p \rightarrow n = V_{in} \frac{2^N}{V_{ref}}$$

Q. A dual slope ADC has the reference voltage, $V_{ref} = 16V$. If the counter's output is 10110101 (in binary) and the clock period is 1.5 MHz, what is the corresponding input voltage and conversion time.

Solⁿ: num of bit $N = 8$, $V_{ref} = 16V$
O/p, $n = \text{decimal of } (10110101)_2$
 $= 181$

$$f_{clk} = 1.5 \times 10^6, T = \frac{1}{f_{clk}}$$

$$\therefore \text{input Voltage, } V_{in} = \frac{n}{2^N} \times V_{ref}$$
$$= \frac{181}{2^8} \times 16$$

$$\boxed{V_{in} = 11.3125V}$$

$$T_1 = 2^N T = \frac{2^N}{f_{clk}} = \frac{2^8}{1.5 \times 10^6} = 0.171 \text{ ms}$$

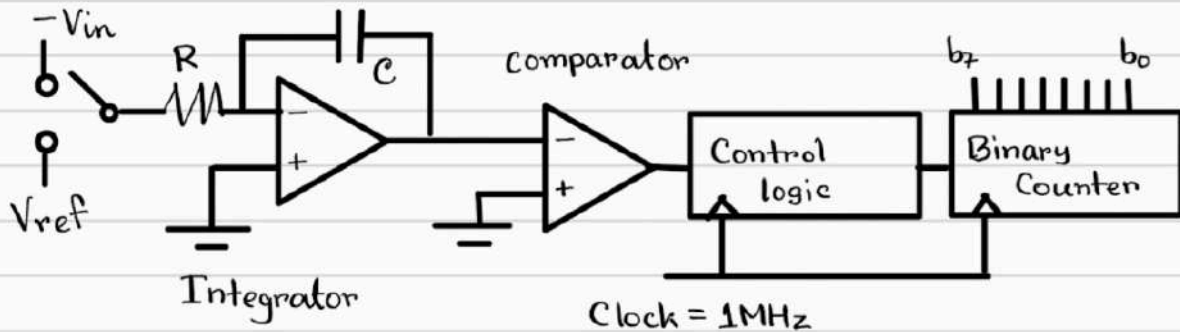
$$T_2 = nT = \frac{n}{f_{clk}} = \frac{181}{1.5 \times 10^6} = 0.121 \text{ ms}$$

$$\therefore \text{Total reading time / Conversion time} = T_1 + T_2$$
$$= 0.292 \text{ ms}$$

(Ans.)

Dual-Slope A/D Converter

- A dual slope A/D converter has $V_{ref} = 5V$, and a 8 bit counter that outputs an 8-bit representation of the input signal. A 1 MHz clock is used for the clock and counter circuits



- ① Determine the input voltage range, number of steps and step size (resolution) of the A/D converter.
- ② For a specific input voltage, the ADC outputs a count of $m=100$. Determine the input voltage.
- ③ Determine the total time required to get the reading in ②
- ④ Calculate the maximum sampling rate of the ADC.
- ⑤ A person wishes to get a sampling rate 4 times that of the current circuit. Determine how many bits should be used for the counter without changing any other system parameters.

(i) Input voltage range : $5V$ ($0 - V_{ref}$)
 number of steps $= 2^n = 2^8 = 256$
 step size $= \frac{\text{voltage range}}{\text{number of steps}} = 0.0195 V$

(ii) We know, $V_{in} = \frac{m}{2^n} V_{ref} = \frac{100}{256} \times 5V = 1.95V$

(iii) For a single conversion, the integrator

(a) integrates $-V_{in}$ for t_1 duration as counter counts from $0 - 2^n$

$$t_1 = \frac{2^n}{\text{clock rate}} = \frac{256}{1 \times 10^6} = 2.56 \times 10^{-4} s$$

(b) integrates V_{ref} for t_2 duration as counter counts from $0 - m$

$$t_2 = \frac{m}{\text{clock rate}} = \frac{100}{1 \times 10^6} = 1 \times 10^{-4} s$$

$$\text{total conversion time} = t_1 + t_2 = 3.56 \times 10^{-4} s$$

(iv) Maximum sampling rate is limited by the time required for worst case conversion of input $= 5V$

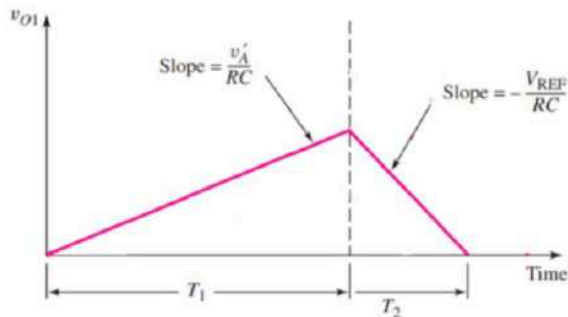
In that case, $t_1 = t_2 = 2.56 \times 10^{-4} s$

thus, worst case conversion duration $= t_1 + t_2 = 5.12 \times 10^{-4} s$

$$\text{Max sampling rate} = \frac{1}{5.12 \times 10^{-4} s} = 1953.125 \text{ Hz}$$

(v) Sampling duration $\propto 2^n$ // Sampling rate $\propto \frac{1}{2^n}$

For 4 times sampling rate, counter bit $= (8 - 2) = 6$



(a)	Find the value of v_{O1} at $t=T_1$, suppose $R=25k\Omega$ and $C=2\mu F$.
(b)	If $R=50k\Omega$ and we $C=2\mu F$, calculate the new value of V_{O1} at $t=T_1$, compare it with (a) and comment on the changes between two output voltages.

$$a) \text{ slope} = \frac{\Delta y}{\Delta x} = \tan \theta = \frac{OB}{AB}$$

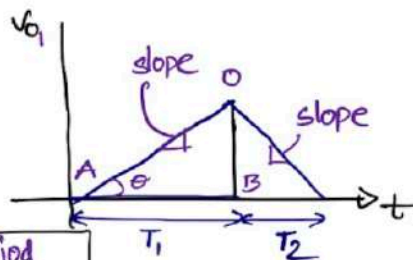
$$\text{As, } \tan \theta = \frac{V'_A}{RC}; \text{ so } \frac{OB}{AB} = \frac{V'_A}{RC}$$

$$\Rightarrow v_{O1} = OB = \frac{V'_A}{RC} \times AB$$

$$\Rightarrow v_{O1} = \frac{V'_A}{RC} \times T_1$$

$$\Rightarrow v_{O1} = \frac{2}{25k \times 2\mu F} \times 2^6 \times 6\mu s$$

$$v_{O1} = 0.01536V$$



$$T_1 = 2^N \cdot 1 \text{ clock period} \\ = 2^N \cdot T = 2^6 \cdot 6\mu s$$

$$b) v_{O1} = \frac{2}{50k\Omega \times 2\mu F} \times 2^6 \times 6\mu s$$

$$v_{O1} = 0.00768V$$

Comment:

As, resistance increases by 2, it decreases the output by half.